

2024 Linear Algebra and Calculus Midterm

① a) $\lim_{x \rightarrow 1} \frac{\sqrt{x}-1}{x-1} = \lim_{x \rightarrow 1} \frac{(\sqrt{x}-1) \cdot (\sqrt{x}+1)}{(x-1)(\sqrt{x}+1)} = \lim_{x \rightarrow 1} \frac{(x-1)}{(x-1)(\sqrt{x}+1)} = \lim_{x \rightarrow 1} \frac{1}{\sqrt{x}+1} = \frac{1}{2}$

b) $\lim_{x \rightarrow \infty} \frac{-3x^4 + 5x^2}{x^3 - x + 7}$. In this case, since it's a rational function whose numerator has a higher degree and thus

will grow faster, the limit will be given by the leading term, taking into account the negative sign:
 $\lim_{x \rightarrow \infty} \frac{-3x^4 + 5x^2}{x^3 - x + 7} = -\infty$. Otherwise, we can divide numerator and denominator by the leading term x^4 :

$$\lim_{x \rightarrow \infty} \frac{-3x^4 + 5x^2}{x^3 - x + 7} = \lim_{x \rightarrow \infty} \frac{-3 + 5/x^2}{1/x - 1/x^3 + 7/x^4} = -\infty.$$

② $f(x) = \frac{\sin x}{x}$. Find $\lim_{x \rightarrow 0} f(x)$.

Since $\cos x \leq f(x) \leq 1$ when x is close to 0 and $\lim_{x \rightarrow 0} \cos x = 1$ and $\lim_{x \rightarrow 0} 1 = 1$, by the Squeeze Theorem,
 $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$.

③ Show $\exists c \in \mathbb{R}$ s.t. $c + \sin c = 1$.

Notice that equation $x + \sin x = 1$ is equivalent to $x + \sin x - 1 = 0$, so the solutions to the equation will be the zeros of the function $f(x) = x + \sin x - 1$. We can check for the images of several points, as $f(0) = -1$, or $f(1) = 1 + \sin 1 - 1 = \sin 1$. Since f is continuous, $f(0) < 0$ and $f(1) > 0$, by Bolzano's Theorem there exists a $c \in (0, 1)$ such that $f(c) = 0$, that is, $c + \sin c = 1$.

④ For f , x is a critical point if $f'(x) = 0$. Find the critical points of $f(x) = \sqrt[3]{x^2}(2x-1)$.

First we need to find $f'(x)$. Knowing that $f(x) = \sqrt[3]{x^2}(2x-1) = x^{2/3}(2x-1)$.

For that we can either use the product rule: $f'(x) = \frac{2}{3}x^{-1/3}(2x-1) + 2x^{2/3} = \frac{4}{3}x^{2/3} - \frac{2}{3}x^{-1/3} + 2x^{2/3} = \frac{10}{3}x^{2/3} - \frac{2}{3}x^{-1/3}$,
 or by expanding the function first: $f(x) = x^{2/3}(2x-1) = 2x^{5/3} - x^{2/3}$, and differentiating second: $f'(x) = \frac{10}{3}x^{2/3} - \frac{2}{3}x^{-1/3}$.
 Now we just need to solve $f'(x) = 0$, so $\frac{10}{3}x^{2/3} - \frac{2}{3}x^{-1/3} = 0 \rightarrow 5x^{2/3} - x^{-1/3} = 0 \rightarrow 5x^{2/3} = x^{-1/3} \rightarrow 5x^{2/3} \cdot x^{1/3} = 1 \rightarrow 5x = 1 \rightarrow x = \frac{1}{5}$

⑤ $\int_0^1 \log x \, dx$.

Let's use integration by parts and the fundamental theorem of calculus. Take $u = \log x \rightarrow du = \frac{1}{x} dx$ and $dv = dx \rightarrow v = x$.

Now $\int_0^1 \log x \, dx = uv \Big|_0^1 - \int_0^1 v \, du = \log x \cdot x \Big|_0^1 - \int_0^1 x \cdot \frac{1}{x} \, dx$. Here we see we're dealing with an improper integral, so we

rewrite it as $\int_0^1 \log x \, dx = \lim_{R \rightarrow 0} \int_R^1 \log x \, dx = \lim_{R \rightarrow 0} \left[x \cdot \log x \Big|_R^1 - \int_R^1 dx \right] = \lim_{R \rightarrow 0} \left[-R \cdot \log R - [x]_R^1 \right] = \lim_{R \rightarrow 0} \left[-R \cdot \log R - [1 - R] \right] =$

$\lim_{R \rightarrow 0} -R \cdot \log R - 1 + R$. Since the function $g(R) = R$ goes to zero faster than $f(R) = \log R$, we have that

$$\lim_{R \rightarrow 0} -R \cdot \log R - 1 + R = -1.$$

⑥ Compute the Jacobian of the function $f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ given by $f(x, y, z) = (e^x e^y, y - x^2 z, z \cos y + x^3)$. Evaluate it at 0.

For that we need to compute all the partial derivatives of each function.

$$J_{f(x, y, z)} = \begin{bmatrix} e^x e^y & e^x e^y & 0 \\ -2xz & 1 & -x^2 \\ 3x^2 & -z \sin y & \cos y \end{bmatrix}. \text{ Now } J_{f(0, 0, 0)} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$